Exploratory Data Analysis Report for Electricity Demand Forecasting

Paper-ready narrative reconstructed from the uploaded EDA.html

Target variable	TOTAL_DEMAND (electricity demand)
Data scope	Daily observations from 2010-01-08 to 2021-03-17; 4,084 records
Source basis	Uploaded EDA.html containing code, outputs, and figures

1. Study objective and dataset overview

This report reconstructs the completed exploratory data analysis from the uploaded EDA.html into a paper-ready narrative for the electricity demand forecasting task. The purpose is not to reproduce notebook code, but to synthesise the observed outputs, figures, and statistical findings into a coherent empirical analysis of the target variable TOTAL_DEMAND.

The dataset contains 4,084 daily observations and 11 original variables, covering the period from 2010-01-08 to 2021-03-17. For EDA focused on modelling relevance, LOCATION and HOLIDAY_NAME were excluded, leaving nine variables for descriptive analysis, visualisation, and correlation assessment.

Item	Description
Sample size	4,084 daily observations
Original variables	11
Variables used in EDA	9 (excluding LOCATION and HOLIDAY_NAME)
Date range	2010-01-08 to 2021-03-17
Forecast target	TOTAL_DEMAND (electricity demand)
Key candidate predictors	T_MAX, T_MIN, T_MEAN, WEEKDAY, IS_HOLIDAY, DEMAND_LAG1, DEMAND_LAG7

2. Data quality assessment

The dataset shows strong structural quality. Missing-value checks indicate that all nine variables retained for EDA have zero missing observations and zero missing ratio. Duplicate-row checks return a count of zero. As a result, the data can move directly into feature analysis and model preparation without requiring imputation or de-duplication.

The DATE field was successfully converted to datetime format and sorted in ascending order, ensuring that the subsequent time-series analysis is based on a properly ordered daily sequence.

Check	Result
Missing values	All EDA variables have 0 missing observations and 0% missing ratio
Duplicate rows	0 duplicated rows
Date field	DATE successfully parsed to datetime and sorted chronologically
Overall integrity	No obvious structural data-quality defects

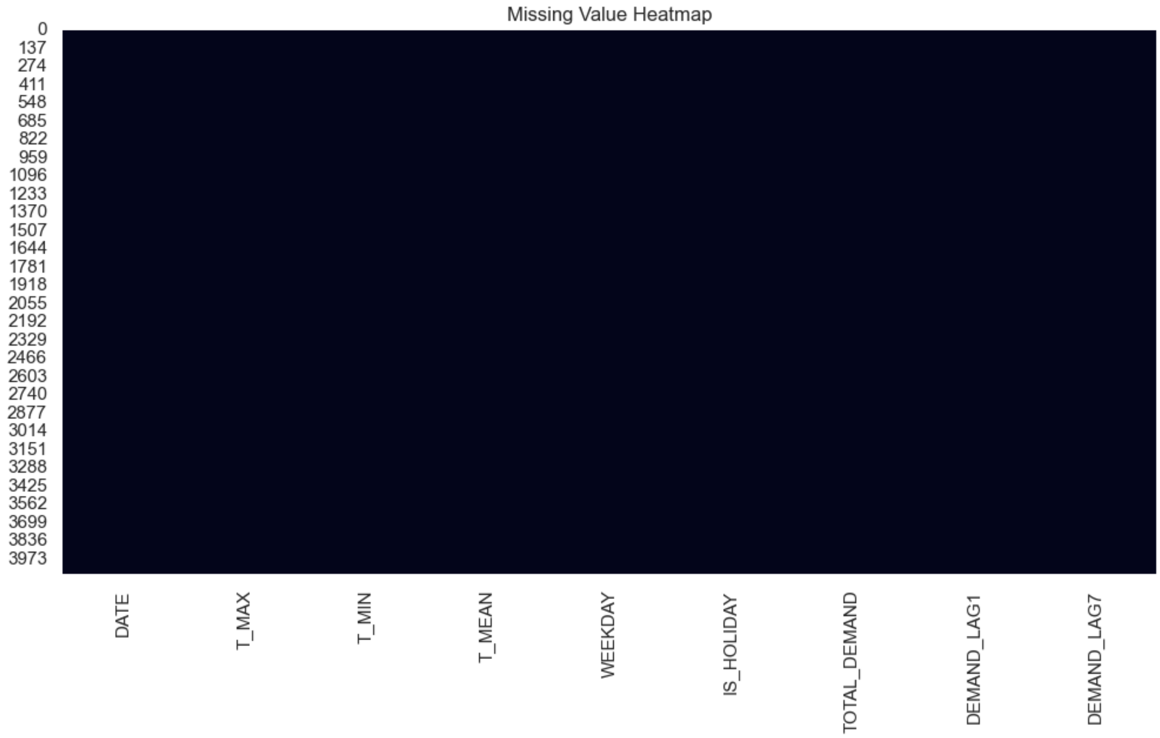

Figure 1. Missing-value heatmap showing no missingness in the analysed variables.

3. Descriptive statistics and target distribution

TOTAL_DEMAND has a mean of 389,412.42, a median of 386,998.83, a standard deviation of 38,359.37, a minimum of 290,102.22, and a maximum of 553,911.24. Its skewness is 0.329 and kurtosis is approximately 0.015, suggesting an approximately unimodal distribution with a mildly extended right tail. In practical terms, the target is not severely skewed, but a small number of high-demand peak days remain visible and may influence regression error distribution.

This pattern implies that the series is suitable for direct modelling without aggressive transformation as a default choice, although peak-demand observations should still be monitored during model evaluation.

Statistic	Value
Mean of TOTAL_DEMAND	389,412.42
Median of TOTAL_DEMAND	386,998.83
Standard deviation	38,359.37
Minimum	290,102.22
Maximum	553,911.24

Skewness	0.329
Kurtosis	0.015

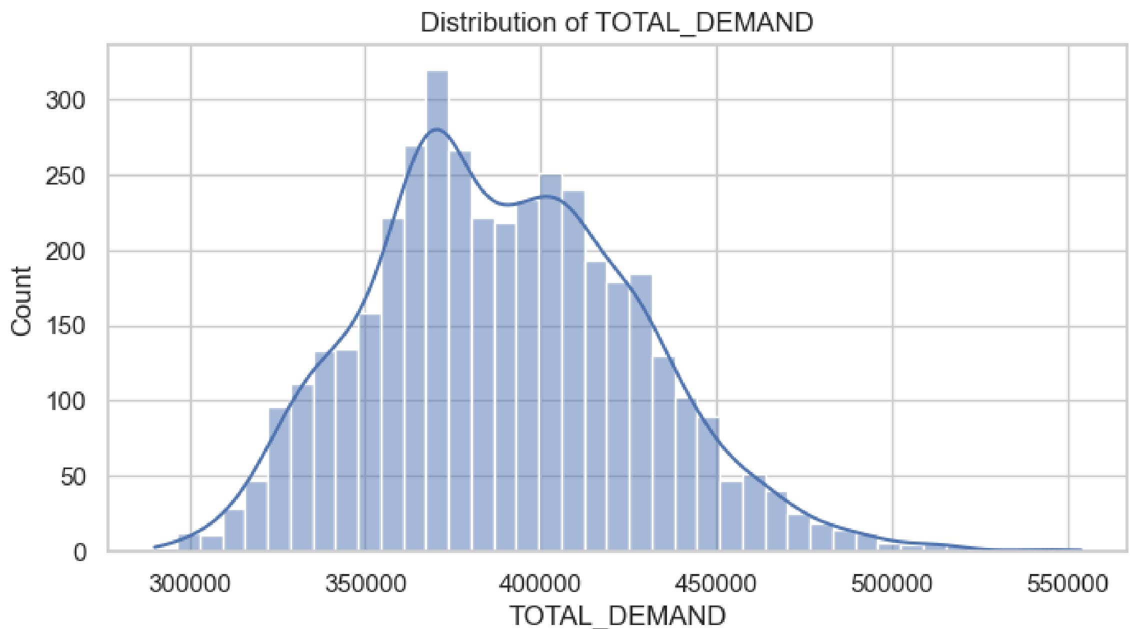

Figure 2. Distribution and boxplot of TOTAL_DEMAND.

Table 1. Selected descriptive statistics for key numeric variables

Variable	Mean	Std. Dev.	Min	Max
T_MAX	22.91	5.05	11.70	44.70
T_MIN	12.49	5.29	-1.30	25.60
T_MEAN	17.40	4.69	4.94	33.31
TOTAL_DEMAND	389,412.42	38,359.37	290,102.22	553,911.24
DEMAND_LAG1	389,413.19	38,346.33	290,102.22	553,911.24
DEMAND_LAG7	389,479.23	38,346.82	290,102.22	553,911.24

4. Temporal trend and seasonality

The daily time-series plot reveals strong temporal dependence and recurring fluctuations in TOTAL_DEMAND. At the long-horizon level, demand appears relatively higher in the early part of the sample and somewhat lower in later years, although substantial volatility remains throughout the full period.

At the seasonal level, repeated peaks and troughs suggest regular cyclic behaviour, consistent with electricity demand being driven jointly by weather conditions, calendar effects, and habitual consumption patterns. The year-wise boxplots indicate cross-year differences in both central tendency and dispersion, while the month-level average-demand curve confirms a clear monthly seasonal component.

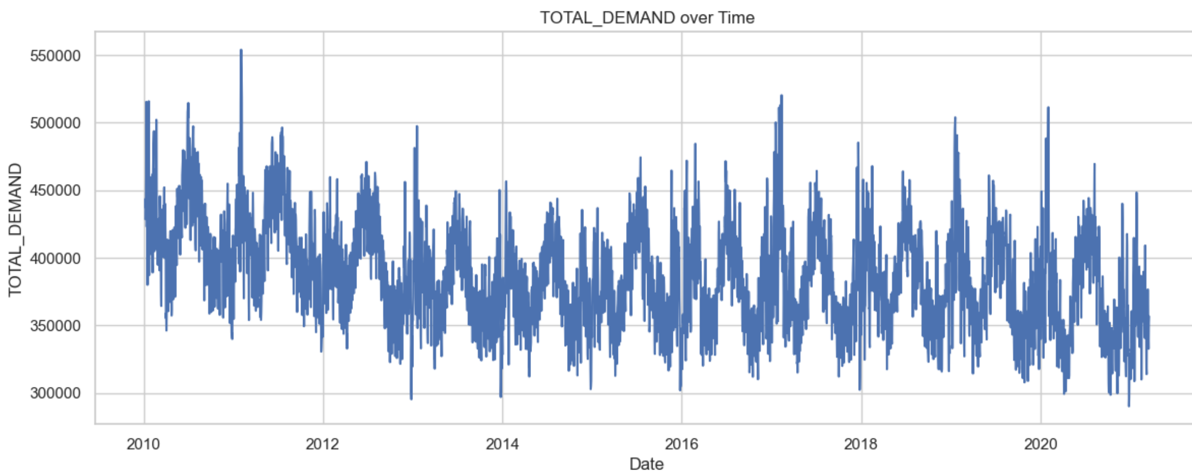

Figure 3. Daily trajectory of TOTAL_DEMAND over the sample period.

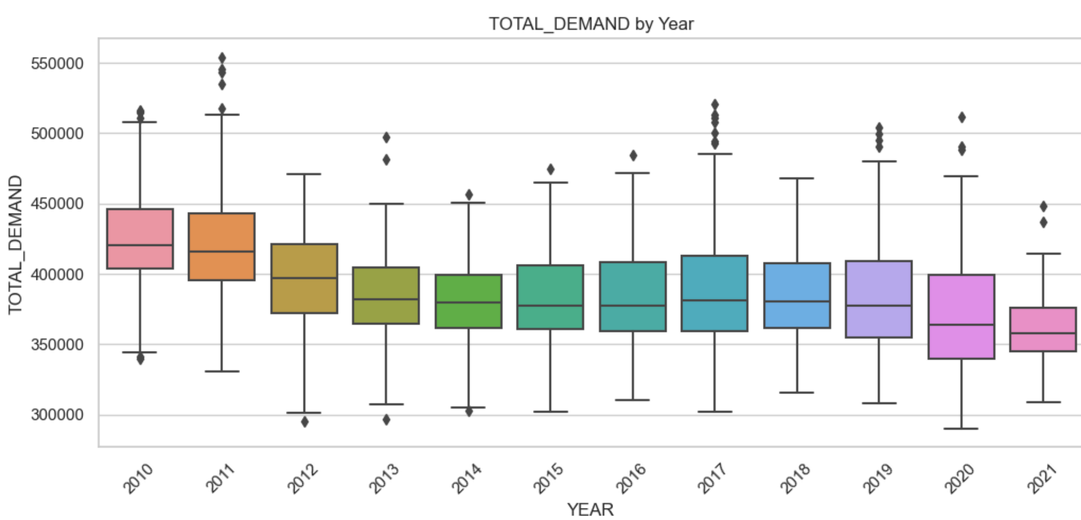

Figure 4. Distribution of TOTAL_DEMAND by year.

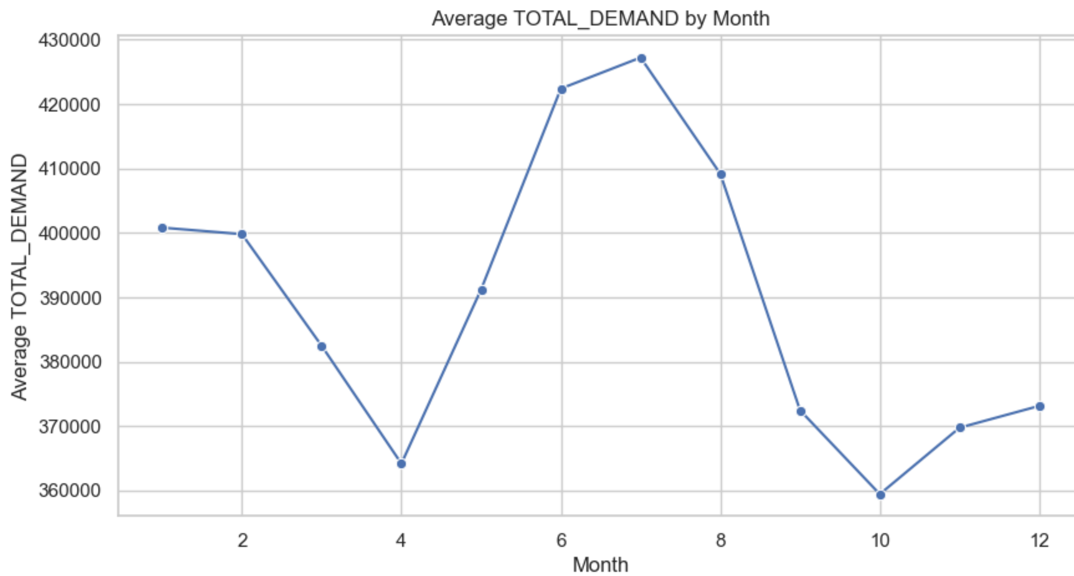

Figure 5. Average TOTAL_DEMAND aggregated by month.

5. Calendar effects: weekday and holiday patterns

The distribution of WEEKDAY is almost perfectly balanced, with each coded category containing approximately 583-584 observations. This is useful because it reduces the risk that the estimated weekday effect is driven by sample imbalance.

Despite the balanced frequency, average demand differs materially across weekday codes. WEEKDAY=2 has the highest average demand at 401,191.72, whereas WEEKDAY=7 is the lowest at 359,915.08. This indicates that intra-week behavioural patterns are economically meaningful for demand forecasting.

The holiday effect is even more pronounced. Average demand on non-holidays is 390,922.34, compared with 343,487.67 on holidays, a difference of about 47,434.67, or roughly 12.1% lower on holidays. Although holiday observations account for only 130 records, or 3.18% of the sample, the direction and magnitude of the effect are both clear.

Table 2. Average demand by weekday code

Weekday code	Average TOTAL_DEMAND
2.0	401,191.72
4.0	400,833.81
3.0	399,591.26
5.0	398,786.43
1.0	395,182.85

6.0	370,332.06
7.0	359,915.08

Table 3. Average demand on holidays vs. non-holidays

Group	Average TOTAL_DEMAND
Non-holiday	390,922.34
Holiday	343,487.67

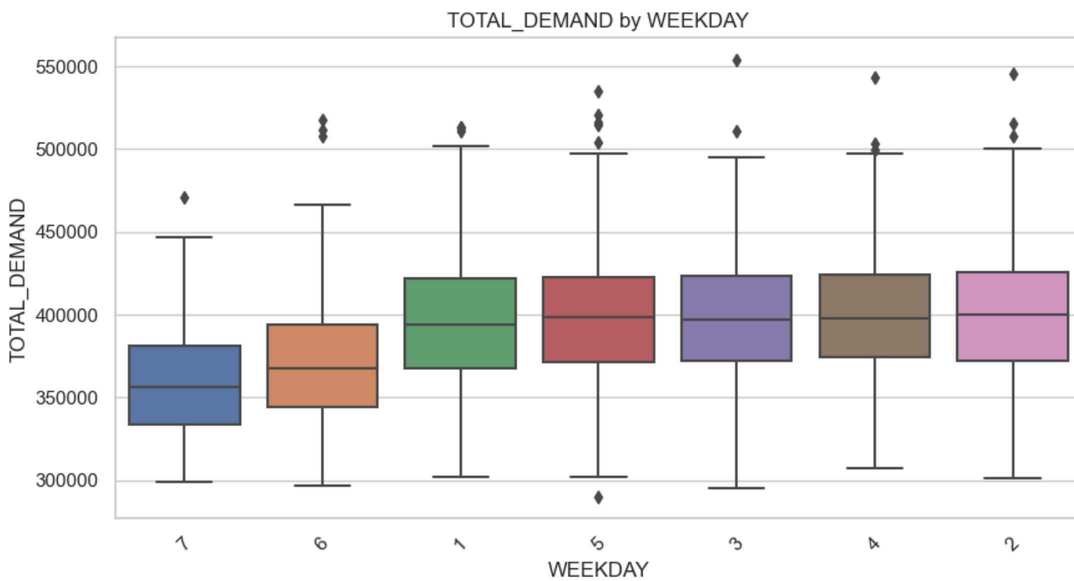

Figure 6. Effect of WEEKDAY on TOTAL_DEMAND.

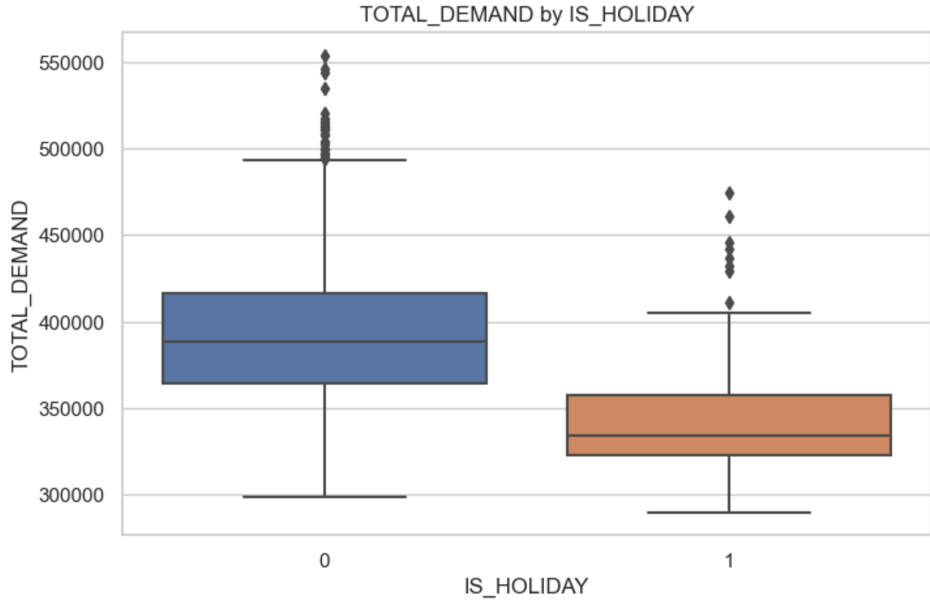

Figure 7. Difference in TOTAL_DEMAND between holidays and non-holidays.

6. Relationships between predictors and the target

Correlation analysis shows that the lag features are by far the strongest linear correlates of TOTAL_DEMAND. The correlation with DEMAND_LAG1 is 0.7994, and with DEMAND_LAG7 it is 0.7154. These are substantially stronger than the correlations of the temperature or calendar variables, indicating that short-run persistence and weekly recurrence are core properties of the demand process.

Temperature variables exhibit weak negative linear correlations with the target: -0.0998 for T_MAX, -0.1548 for T_MIN, and -0.1641 for T_MEAN. This should not be overinterpreted as evidence that temperature is unimportant. A more plausible interpretation is that the temperature-demand relationship is nonlinear and potentially interacts with seasonality and calendar structure.

WEEKDAY and IS_HOLIDAY also matter, with correlations of -0.3134 and -0.2171, respectively. These values are smaller than those of the lag variables, but still strong enough to justify explicit retention in downstream modelling.

Table 4. Summary of the relationship between each predictor and TOTAL_DEMAND

Variable	Type	Missing count	Missing ratio (%)	Correlation with target
DEMAND_LAG1	float64	0	0.0	0.7994
DEMAND_LAG7	float64	0	0.0	0.7154
T_MAX	float64	0	0.0	-0.0998
T_MIN	float64	0	0.0	-0.1548

T_MEAN	float64	0	0.0	-0.1641
IS_HOLIDAY	int64	0	0.0	-0.2171
WEEKDAY	int64	0	0.0	-0.3134

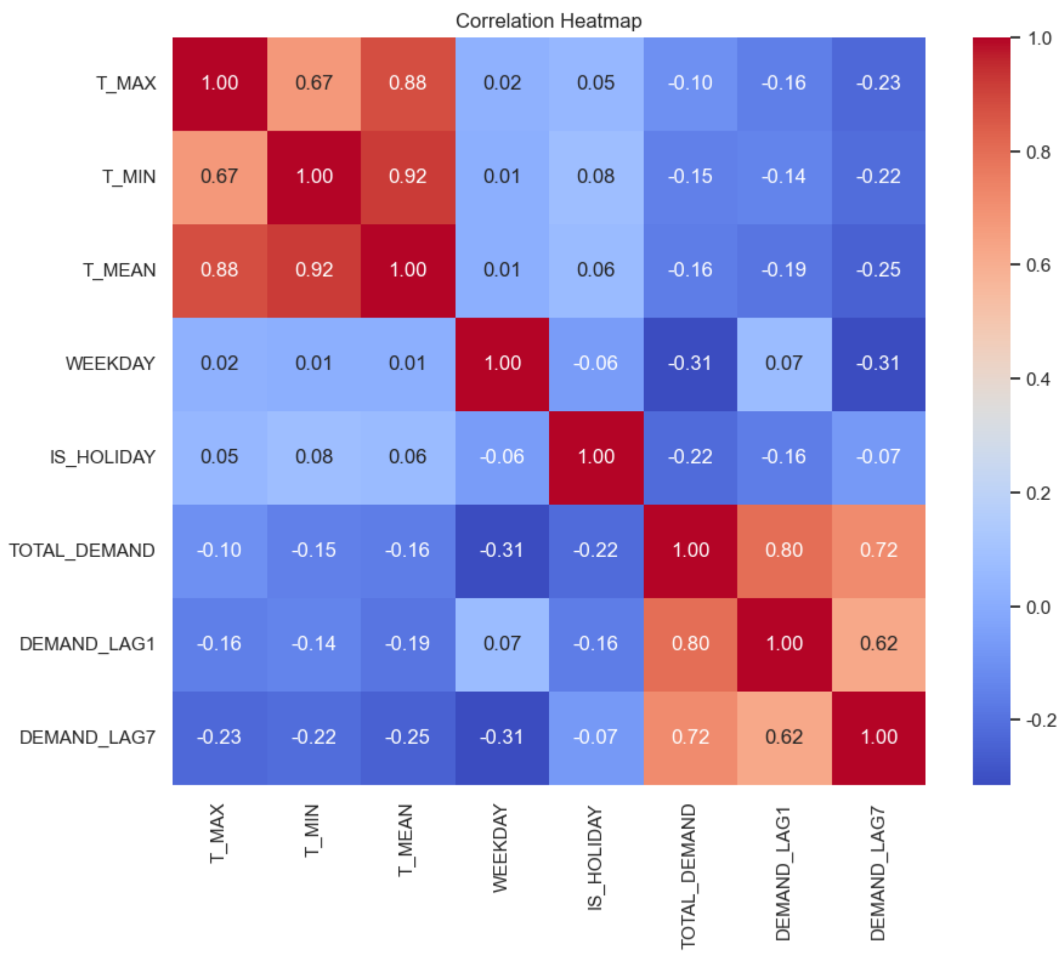

Figure 8. Correlation heatmap across numeric variables.

7. Lag structure and autocorrelation

The dedicated lag analysis reinforces the conclusion that demand is strongly time-dependent. The scatterplot of TOTAL_DEMAND against DEMAND_LAG1 forms a tight positive band, indicating that adjacent days have highly persistent demand levels. The relationship with DEMAND_LAG7 is slightly weaker but still pronounced, consistent with a weekly repeating demand cycle.

The short-lag ACF indicates that TOTAL_DEMAND remains significantly positively autocorrelated across several low-order lags, with the strongest dependence observed at small lags, suggesting strong short-term persistence in electricity demand. The long-lag autocorrelation plot further shows an oscillatory decay pattern over a much wider lag range, indicating that the series contains not only short-term dependence but also sustained seasonal or cyclical structure. This has direct modelling implications: lagged inputs, rolling statistics, seasonal indicators, or explicitly time-aware forecasting models should be prioritised. A model that ignores the temporal structure and relies only on same-day weather and calendar inputs would likely leave a large share of predictive signal unused.

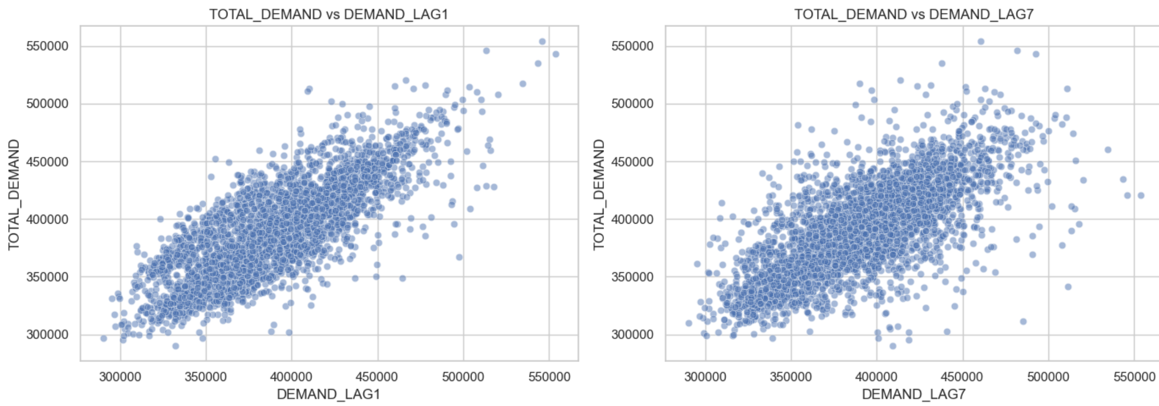

Figure 9. Relationship of TOTAL_DEMAND with DEMAND_LAG1 and DEMAND_LAG7.

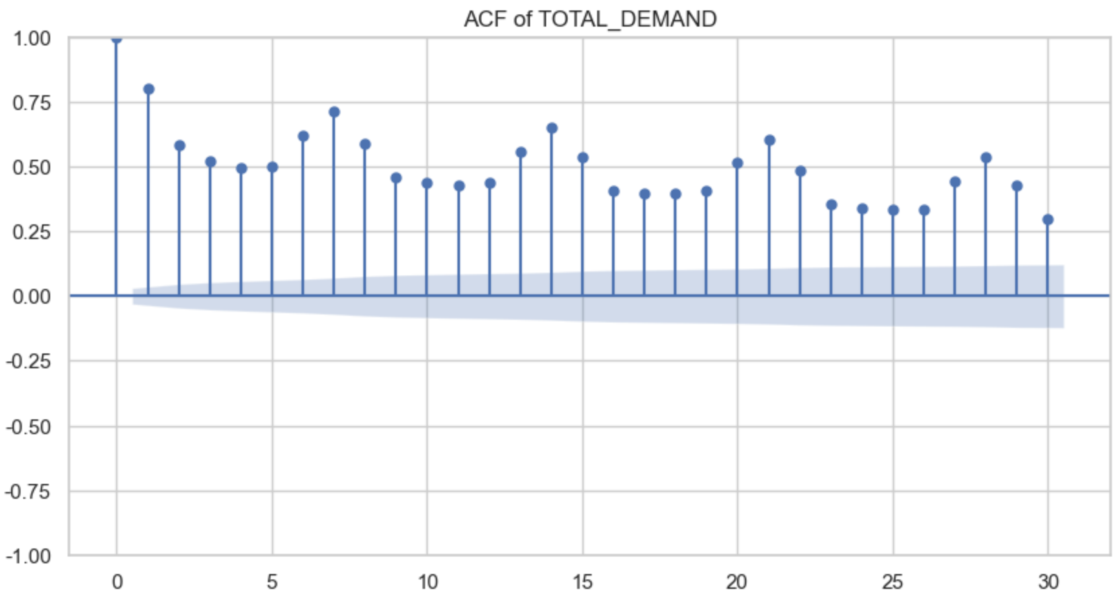

Figure 10. Short-lag ACF of TOTAL_DEMAND.

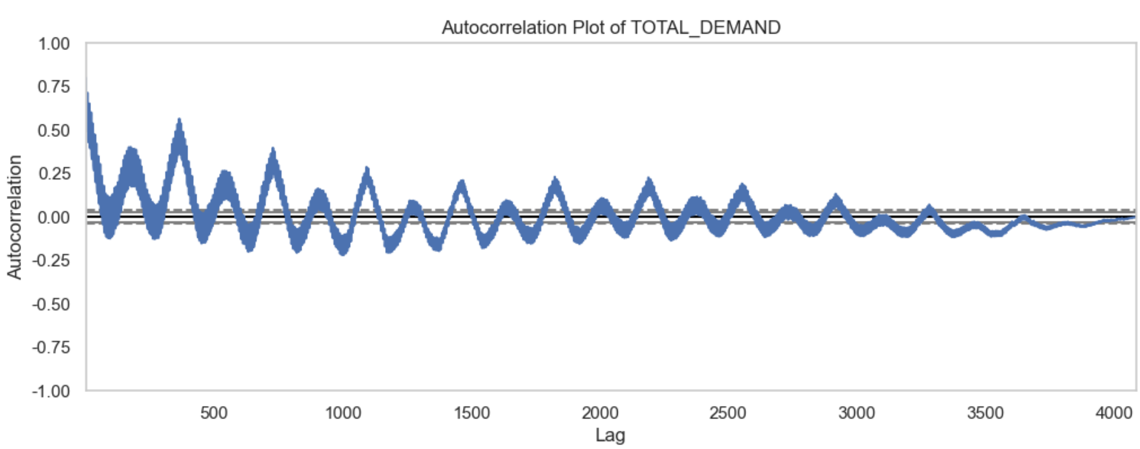

Figure 11. Autocorrelation plot of TOTAL_DEMAND.

8. Temperature multicollinearity and outliers

The three temperature measures are strongly correlated with one another. The correlation between T_MAX and T_MEAN is 0.8786, between T_MIN and T_MEAN is 0.9189, and between T_MAX and T_MIN is 0.6720. This indicates substantial multicollinearity and suggests that the temperature variables carry overlapping information.

For modelling, this matters especially for linear and other coefficient-sensitive methods. Possible responses include variable selection, regularisation, or the construction of more interpretable derived temperature features.

Outlier ratios identified by the IQR rule are modest overall. TOTAL_DEMAND, DEMAND_LAG1, and DEMAND_LAG7 each show an outlier ratio of about 0.66%, while T_MAX has an outlier ratio of about 1.03%. Because such observations may correspond to extreme weather or operational stress periods, they should be handled carefully rather than removed mechanically.

Table 5. Correlations among temperature variables

T_MAX	T_MIN	T_MEAN
1.0000	0.6720	0.8786
0.6720	1.0000	0.9189
0.8786	0.9189	1.0000

Table 6. Outlier summary based on the IQR rule

Variable	Outlier count	Outlier ratio (%)
IS_HOLIDAY	130	3.183
T_MAX	42	1.028
TOTAL_DEMAND	27	0.661
DEMAND_LAG1	27	0.661
DEMAND_LAG7	27	0.661
T_MEAN	1	0.024
T_MIN	0	0.000
WEEKDAY	0	0.000

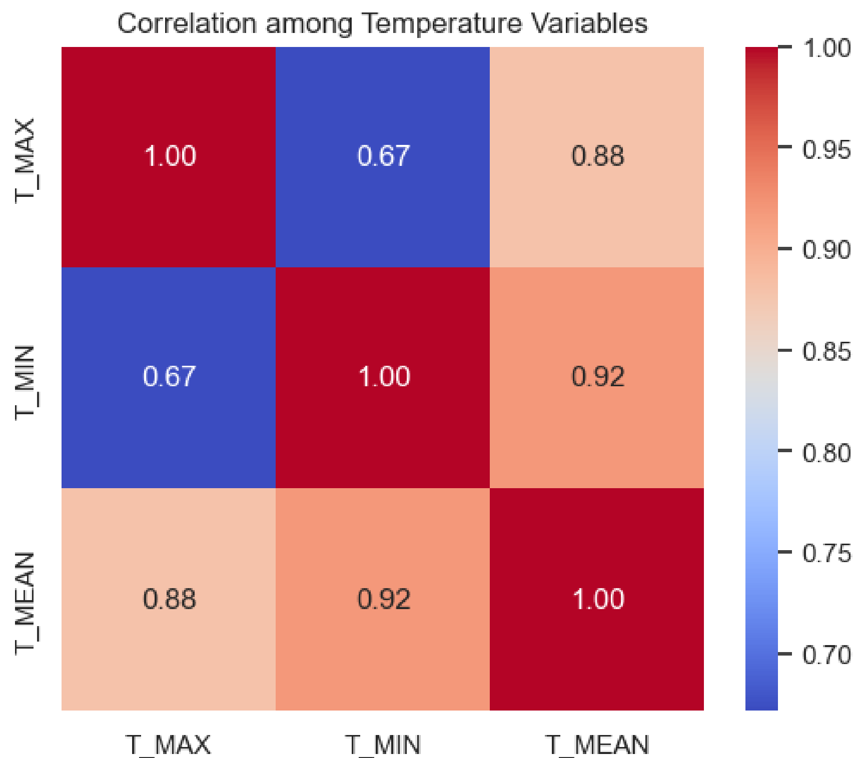

Figure 12. Correlation heatmap among temperature variables.

9. Interaction patterns and modelling implications

The joint month-holiday view indicates that the magnitude of the holiday effect may vary across months, suggesting that holiday impacts are seasonally modulated rather than perfectly constant. Likewise, the weekday-holiday cross-tabulation shows that holiday demand remains lower than non-holiday demand for every weekday code, indicating a robust holiday effect across the weekly cycle.

Taken together, the EDA supports four modelling implications. First, lag variables should be treated as core predictors. Second, temperature effects are likely nonlinear and collinear, so flexible feature engineering or regularised modelling is advisable. Third, WEEKDAY and IS_HOLIDAY should be retained as key calendar variables. Fourth, rare peak-demand observations should be handled through robust modelling and targeted evaluation, not by simplistic deletion.

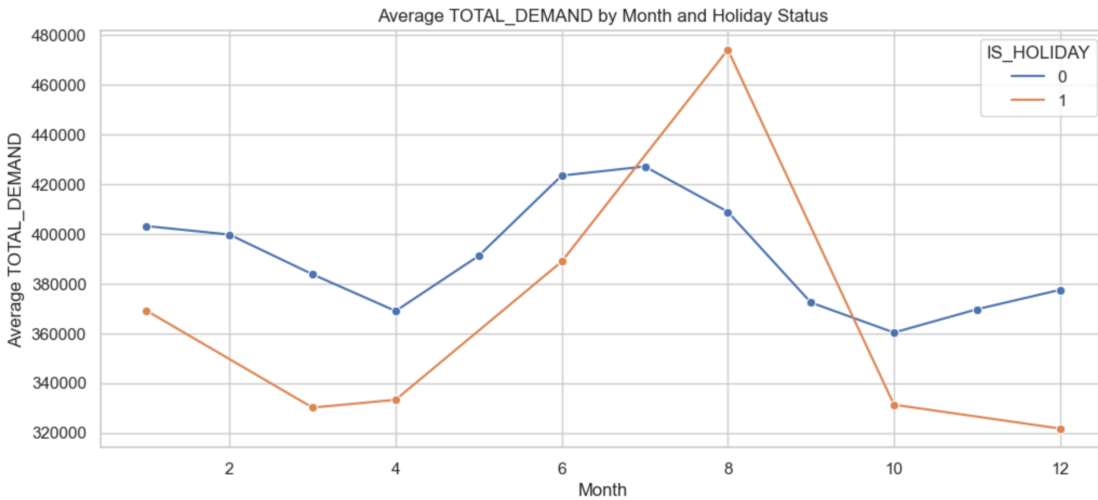

Figure 13. Joint effect of month and holiday status.

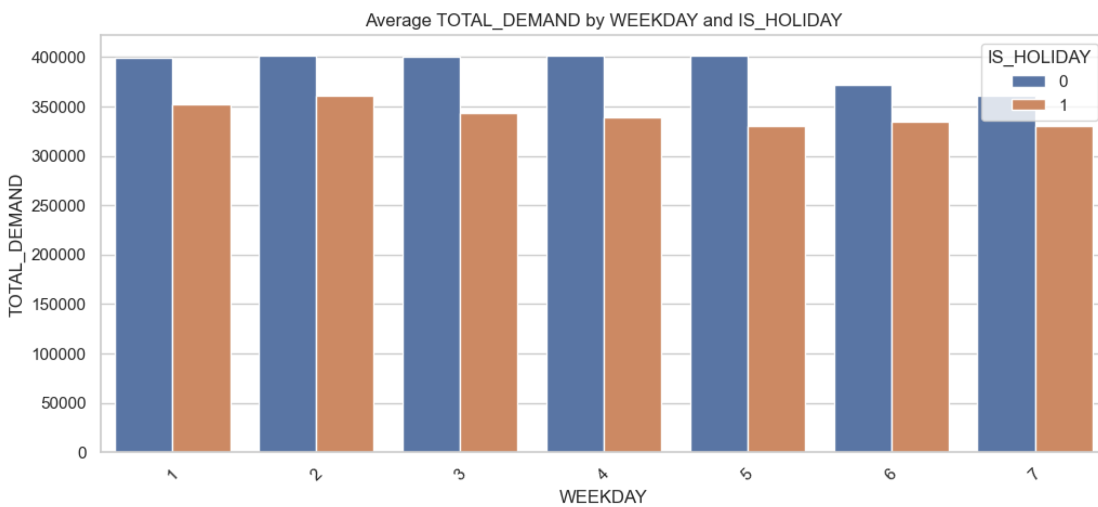

Figure 14. Joint effect of weekday code and holiday status.

10. Conclusion

The EDA shows that TOTAL_DEMAND is a daily electricity-demand series characterised by strong temporal dependence, clear weekly cyclicality, and meaningful calendar effects. The dataset is clean, with no missing values and no duplicate rows. The target distribution is broadly stable but includes a small number of high-demand peaks. Lag variables are the strongest predictors, holidays substantially depress demand, weekday effects are systematic, and the temperature variables are highly collinear.

Accordingly, the downstream forecasting model should explicitly exploit temporal memory, calendar structure, and weather information, while also being evaluated carefully on both normal-demand days and peak-demand episodes.